

EE1101: Circuits and Network Analysis - Quiz 03

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1. The exam duration is 1 hour, starting at 12:10. Ensure that you submit both the question paper and the answer book by 13:10. Please be aware that **late submissions** may incur **penalties**.
2. Read all questions carefully. Misinterpretations will **not** be considered. It is essential to write your solutions clearly and legibly in the answer book. In a previous quiz, a few students submitted answers that were not clearly written. Going forward, such responses will **not** be evaluated.
3. Any form of unfair practice will result in an **F grade** and will be reported to the disciplinary committee.

1. (4 points) Consider the circuit shown in Fig. 1. Assume that the circuit is initially at rest and t_0 is large enough for the circuit to reach steady state before the switching occurs. Clearly write down the differential equations that govern the response by choosing the references as shown in Fig. 1.

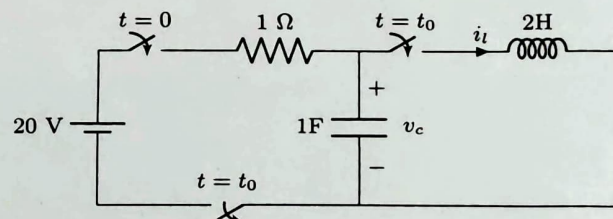


Fig. 1: Circuit for Question 1

- (a) (1 point) Determine the expression for $v_c(t)$ for $0 \leq t < t_0$.
 - (b) (2 points) Determine the expression for $v_c(t)$ for $t \geq t_0$.
 - (c) (1 point) Determine the expression for $i_l(t)$ for $t \geq t_0$.
2. (3 points) Consider the circuit shown in Fig. 2. Assume that the circuit is initially at rest. Determine the value of t_0 for which the transient response of $v_c(t)$ for $t \geq t_0$ is zero.

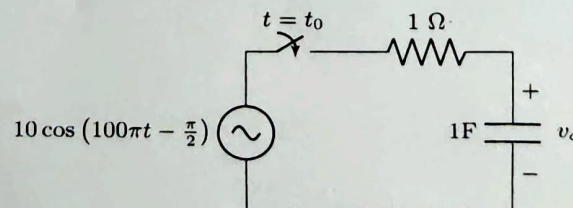


Fig. 2: Circuit for Question 2

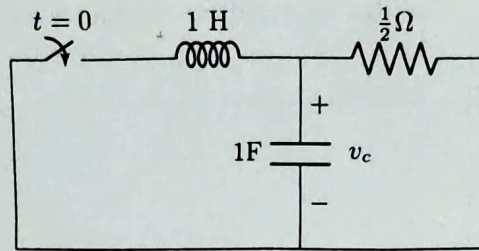


Fig. 3: Circuit for Question 3

3. (3 points) Consider the circuit shown in Fig. 3. Assume that $i_L(0-) = 10\text{ A}$ and $v_c(0-) = 0\text{ V}$.
- (a) (1.5 points) Determine the expression for $v_c(t)$ for $t \geq 0$.
 - (b) (1.5 points) Determine the expression for $i_L(t)$ for $t \geq 0$.